



DataCo Global

Phase 1 | Problem Definition & Project Scope

Project	Supply Chain Analytics Improvement
Business Problem	55% late delivery rate impacting customer satisfaction

1. Executive Summary

DataCo Global is a multi-channel retail company operating across international markets. Analysis of supply chain data indicates approximately 55% of orders are flagged with late delivery risk, significantly above industry benchmarks for comparable retail operations.

This project aims to analyze the supply chain operations, identify root causes of delivery delays, and provide data-driven recommendations to improve on-time delivery performance while maintaining operational efficiency.

2. Business Context

DataCo Global operates in the retail sector with customers across multiple countries and market regions. The company processes orders through an ERP system that captures end-to-end transaction data from order placement through delivery.

Key Business Facts

- Multi-channel retailer serving Consumer, Corporate, and Home Office segments.
- Global operations with customers across US, EU, LATAM, and APAC markets.
- E-commerce presence supported by clickstream behavior tracking.
- Product catalog organized across departments and categories.

Available Data

Two datasets are available for analysis: a primary transaction dataset containing order, customer, product, and shipping records, and a supplementary clickstream dataset capturing web browsing behavior. An official data dictionary file accompanies the transaction dataset.

3. Problem Statement

The company's supply chain shows a late delivery rate of approximately 55%. This level of service failure has direct business implications:

- Customer satisfaction and retention risk due to missed delivery promises.
- Increased customer service overhead from delivery-related inquiries.
- Potential revenue impact from repeat customer attrition.
- Operational inefficiency from expedited shipping or re-shipments.

The root cause is unknown. Possible contributing factors include carrier performance, geographic coverage gaps, inaccurate scheduled delivery windows, or systematic issues in order fulfillment processes. This analysis will identify which factors drive the majority of late deliveries.

4. Project Objectives

Objective	Description	Success Criteria
Data Profiling	Assess data quality, completeness, and consistency across datasets.	Completeness and quality scorecard produced
Delivery Analysis	Quantify late delivery patterns by region, product, and customer segment.	Root cause factors identified and ranked
Performance Metrics	Calculate on-time delivery rates, average delay durations, and cost impact.	Baseline KPIs established for monitoring
Recommendations	Provide actionable recommendations to improve delivery performance.	Prioritized improvement roadmap delivered

5. Scope

In Scope

- Analysis of transaction data from DataCoSupplyChainDataset.csv (53 fields, ~180K rows).
- Analysis of clickstream behavior data from tokenized_access_logs.csv (8 fields, ~415K rows).
- Customer, product, order, shipping, and financial data domains.
- Historical data covering orders from 2015–2018 (inferred from date ranges).

Out of Scope

- Real-time data integration or system implementation.
- External data sources (carrier data, weather, traffic).
- Financial audit or reconciliation.
- System architecture design or technology selection.

6. Stakeholders

The following business functions are expected to have an interest in this analysis. Specific stakeholder names and organizational reporting structures are not known.

Business Function	Interest in Analysis	Expected Use of Findings
Logistics / Supply Chain Operations	Understanding delivery delay patterns and carrier performance.	Operational improvements and carrier management.
Customer Experience / Service	Impact of late deliveries on customer satisfaction.	Service level improvements and communication strategies.
Sales & Marketing	Customer segmentation and retention strategies.	Targeted campaigns and segment-level offers.
Finance	Cost impact of late deliveries and operational changes.	ROI analysis and budget allocation for improvements.
Merchandising / Product	Product categories with highest delay rates.	Assortment and inventory placement decisions.

7. Success Metrics

Metric	Current State (Inferred)	Target Outcome
Late Delivery Rate	~55% of orders flagged at risk	Root causes identified and quantified
Data Completeness	TBD from profiling phase	Known data quality gaps documented
Analysis Coverage	No prior analysis documented	Full diagnostic completed across all domains
Recommendations	None	Prioritized action plan delivered

Current state metrics are inferred from available data. Baseline values will be refined during the Data Profiling phase.

End of Business Requirement Document. Proceed to Phase 2: Operational Architecture Discovery.